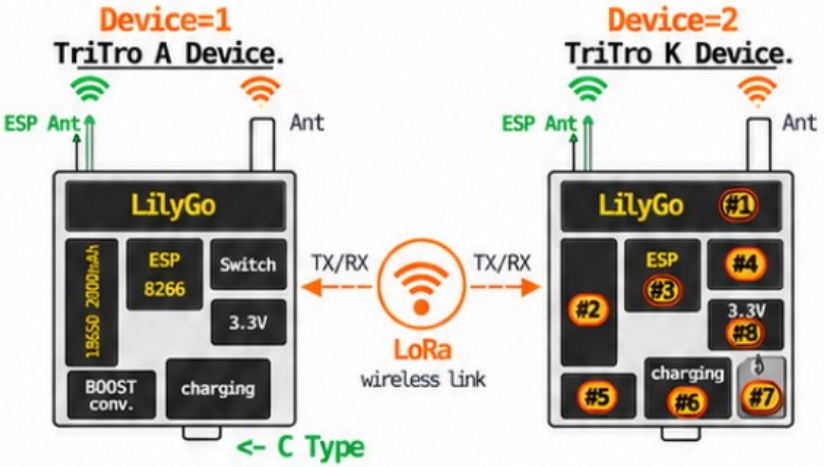
# LoRa Communication V2V & chat offline.



## TriTro A K Devices.




## Component List

- #1 LilyGo Module
- #2 Lithium Battery 3.7V 2000mAh
- #3 ESP (ESP8266)
- #4 Switch
- #5 Boost Converter 3.7V->5V
- #6 C-Type Charging Module
- #7 PCB Board (all parts mounted)
- #8 Buck-Boost Converter 3.7V->3.3V

## Components used for

- #1 → Lilygo Module.
- #2 → Lithium Battery 3.7V 2000mAh
  - To power > Lilygo -> 5V
    - ESP -> 3.3V
  - #5 (3.7 input) → Boost Converter-5V o/p -> lilygo- #1
  - #8 (3.7 input) → 3.3V Buck Booster Converter
    - 3.3V o/p -> ESP- #3
    - Switch-> #4 → lilygo- #1
    - #2 → ESP- #3
    - #8
  - #6 → C Type charging Module
    - input 5V from C-Type charger.



## ▲ Process To start The Both devices

↳ for the Both A & K device Same Procedure should be followed.

# -> Switch "on" the push Button Switch

↳ Switch -> #4

=> After Switching on the #4 the ESP Module gets started ..... !

First we see **how to start** and **communicate** LilyGo.

↳ #4 Keep "ON"

↳ In the LilyGo board there is an "slide switch"

! It Should be always in the "OFF" state unless you use voice to voice Communication

↳ slide switch -> turn "ON"

### OLED Display

```
TriTro A or K.
Ready.
RX Mode
Listening....
BOOT = Tx
```

Display shows this type of information. It tells you the Device in RX or TX.

↳ RX -> listen from TX.

↳ TX -> you should speak on this.

↳ for V2V -> Press Boot mode on Both Devices to switch RX/TX.

## # How does chat system work on phone offline ?



=> ESP -> with given uploaded code file names

Device=1 -> offline chat TriTro A Code

Device=2 -> offline chat TriTro K Code

### "App Set up"

=> offline web application "Enable HTTP" Flag.

Code -> inbuilt web application

### Steps

1) -> open chrome
2) -> Type in the address bar => chrome://flags
3) -> Press Enter.
4) -> in the Search box, Type => insecure
5) -> Find "Insecure Origins Treated as Secure"
6) -> In the Text box below it, Type: http://192.168.4.1
7) -> change the dropdown from "Default" To "Enabled" Tap the "Relaunch" button at the bottom.
8) -> After chrome restarts, try to install by typing 192.168.4.1 in the chrome search bar. Download or Install -> click "Install".

Settings -> Search "Download" or "Install"

application -> Now Install should work.

**Application Interface in Home Screen**

ECE → icon on Home Screen

## # Now How To chat.



### Steps

1) -> Keep the ESP in the ON state only.
2) -> "Turn off Sim, Internet everything related to Internet. in the Mobile phone."
3) -> open wifi Settings.
   - ↳ Turn ON.
   - ↳ shows our device Name Example { TriTro A or TriTro K }
4) -> click which device you want to pair in holding mobile.
5) -> it Asks for password. which is 12345678
   - ↳ final display of your wifi Settings. Search bar:

SSID -> TriTro A
password -> 12345678 — Device #1

SSID -> TriTro K
password -> 12345678 — Device #2

Now open the application installed before. it shows 100% loading.

↳ click "Start" -> Next -> "chat offline"

# Circuit Diagram

LilyGO module + ESP8266 powered from one 3.7V Li-ion battery (TP4056 charging)

